

LOVELY PROFESSIONAL UNIVERSITY

Sample Question Set - Unit 1

Course Code: CSE276

Course Title: ARTIFICIAL INTELLIGENCE
FOUNDATIONS

Course Outcome: CO1

Paper: Sample Multiple Choice Questions

Maximum Marks: 40 x 1 = 40

Unit: 1

Unit Title: Foundations of AI and AI Problem Solving

Unit Mapped RBT: L2 (Understand)

Total Questions: 40

Suggested Duration: 60 Minutes

Answer Grid (to be filled by the student)

Q. No.	Answer	Q. No.	Answer	Q. No.	Answer	Q. No.	Answer
Q1		Q11		Q21		Q31	
Q2		Q12		Q22		Q32	
Q3		Q13		Q23		Q33	
Q4		Q14		Q24		Q34	
Q5		Q15		Q25		Q35	
Q6		Q16		Q26		Q36	
Q7		Q17		Q27		Q37	
Q8		Q18		Q28		Q38	
Q9		Q19		Q29		Q39	
Q10		Q20		Q30		Q40	

Multiple Choice Questions (40 x 1 = 40 Marks)

Q1. Name the capability class of Artificial Intelligence that describes a system competent at exactly one task.

- A) Artificial Narrow Intelligence
- B) Artificial General Intelligence
- C) Artificial Super Intelligence
- D) Artificial Reactive Intelligence

Q2. Identify the functionality class of a system that uses recent past observations as well as the present one, such as a self-driving car.

- A) Reactive machine
- B) Limited memory
- C) Theory of mind
- D) Self-aware

Q3. State the component of a problem formulation that decides whether a state is a solution.

- A) The initial state
- B) The path cost
- C) The goal test
- D) The transition model

Q4. Name the quantity that reports the average number of successors a state has.

- A) The path cost

- B) The solution depth
- C) The state count
- D) The branching factor

Q5. Identify the performance measure that reports whether a search returns the cheapest of the available solutions.

- A) Completeness
- B) Optimality
- C) Time complexity
- D) Space complexity

Q6. Name the field that combines statistics, programming and domain knowledge to obtain insight from data.

- A) Deep learning
- B) Expert systems
- C) Robotics
- D) Data science

Q7. State the phase of the Artificial Intelligence lifecycle in which the performance of a running system is tracked.

- A) Monitoring
- B) Problem definition
- C) Data preparation
- D) Model development

Q8. Identify the capability class that would match a human being across arbitrary tasks.

- A) Artificial Narrow Intelligence
- B) Reactive machine intelligence
- C) Artificial General Intelligence
- D) Limited memory intelligence

Q9. Name the component of a problem formulation that fixes where the search begins.

- A) The goal test
- B) The action set
- C) The initial state
- D) The path cost

Q10. State the performance measure that reports how much memory a search algorithm requires.

- A) Space complexity
- B) Time complexity
- C) Completeness
- D) Optimality

Q11. Identify the subfield that uses multi-layer neural networks to learn its own features from raw input.

- A) Expert systems
- B) Data science
- C) Symbolic reasoning
- D) Deep learning

Q12. Name the function that records the total cost of a sequence of actions.

- A) The heuristic function
- B) The path cost
- C) The goal test
- D) The branching factor

Q13. State the sector in which Artificial Intelligence is applied to yield prediction and to pest identification from leaf photographs.

- A) Manufacturing
- B) Finance
- C) Education
- D) Agriculture

Q14. Name the set of every state that can be reached from the initial state by applying actions.

- A) The solution space
- B) The goal set
- C) The state space
- D) The frontier

Q15. Identify the phase of the Artificial Intelligence lifecycle in which the raw data is cleaned and made ready for use.

- A) Deployment
- B) Data preparation
- C) Monitoring
- D) Problem definition

Q16. State the sector in which Artificial Intelligence is applied to read chest X-rays in support of a radiologist.

- A) Healthcare
- B) Smart cities
- C) Agriculture
- D) Education

Q17. Explain why deep learning is described as a subset of machine learning.

- A) It was invented before machine learning was
- B) It is one family of methods for learning from data
- C) It requires no data of any kind to be trained
- D) It applies only to problems written as rules

Q18. Distinguish a reactive machine from a limited-memory system.

- A) A reactive machine keeps no record of earlier input
- B) A limited-memory system keeps no record of input
- C) Both of the two keep a full record of all input
- D) Neither of the two responds to its current input

Q19. Describe what a performance measure contributes to the comparison of two search algorithms.

- A) A guarantee that both will find a solution
- B) A reduction in the size of the state space
- C) A heuristic estimate of the remaining cost
- D) A common basis on which their behaviour can be judged

Q20. Interpret the following as a functionality class: a chess program that evaluates only the position in front of it and keeps no record of the moves already played.

- A) A limited-memory system
- B) A theory-of-mind system
- C) A reactive machine
- D) A self-aware system

Q21. Predict the effect on an exhaustive search of increasing the depth of the shallowest solution by one.

- A) The number of states examined is multiplied by the branching factor
- B) The number of states examined is increased by one

- C) The number of states examined is left unchanged
- D) The number of states examined is halved

Q22. Recognize the application area described: sensors and cameras adjust the timing of traffic signals across a city so that waiting time falls.

- A) Healthcare
- B) Manufacturing
- C) Smart cities
- D) Education

Q23. Explain why a problem has to be well defined before any search technique can be applied to it.

- A) The algorithm needs a labelled training dataset
- B) The algorithm needs states, actions, a goal test and costs
- C) The algorithm needs the answer to be known first
- D) The algorithm needs the problem to be very small

Q24. Summarize the difference between Artificial Intelligence and data science.

- A) Data science builds systems that act, AI produces insight
- B) The two terms describe exactly the same activity
- C) AI uses data while data science never uses data
- D) AI builds systems that act, data science produces insight

Q25. Distinguish a decomposable problem from one that is not.

- A) A decomposable problem cannot be split at all
- B) A decomposable problem has exactly one solution
- C) A decomposable problem splits into independent subproblems
- D) A decomposable problem needs no goal test

Q26. Explain the difference between a problem whose answer is a state and one whose answer is a path.

- A) One needs a heuristic, the other needs none at all
- B) One has a goal test, the other has no goal test
- C) One is deterministic and the other is stochastic
- D) One needs only the final configuration, the other the steps

Q27. Interpret a problem in which the same action taken in the same state can lead to different outcomes.

- A) Its outcome is not predictable with certainty
- B) Its state space must be infinitely large
- C) Its goal test cannot be written down
- D) Its solution must always be a single state

Q28. Describe what the solution space of a problem contains.

- A) Every state reachable from the initial state
- B) Every state or path that satisfies the goal test
- C) Every action available at the initial state
- D) Every heuristic that could be applied to it

Q29. Explain why Artificial General Intelligence has not yet been achieved.

- A) No computer is fast enough to run any AI method
- B) No organisation has attempted to build such a system
- C) No mathematical definition of intelligence exists
- D) No system transfers competence across arbitrary domains

Q30. Recognize the application area described: an assistant builds a personalised revision plan from a student's assessment record.

- A) Finance
- B) Education
- C) Robotics
- D) Agriculture

Q31. Distinguish a machine learning system from a rule-based one by the origin of the rules each uses.

- A) A rule-based system infers its rules from data
- B) Both of the two infer their rules from data
- C) Learning infers the rules, a rule-based system is given them
- D) Both of the two are given their rules by hand

Q32. Explain why the way a problem is represented can matter more than the algorithm chosen to solve it.

- A) A good representation removes most of the search
- B) A representation fixes the branching factor at one
- C) Every algorithm behaves identically on any problem
- D) The representation decides the memory available

Q33. Predict the difficulty of applying Artificial Intelligence in a sector that records very little data.

- A) Learned methods struggle, so rules or expert input are needed
- B) Learned methods perform better on small datasets
- C) The sector cannot use any form of automation
- D) The lifecycle no longer requires a monitoring phase

Q34. Summarize what the characteristics of a problem tell a designer before a method is chosen.

- A) How many people will use the finished system
- B) Which programming language should be used

- C) How much the finished system will cost to run
- D) Whether it can be decomposed, undone and predicted

Q35. Describe what makes a task better suited to Artificial Intelligence than to ordinary programming.

- A) The task has to be completed very quickly
- B) The rules cannot be written down in advance
- C) The task involves a large volume of storage
- D) The task must run on specialised hardware

Q36. Explain how Artificial Intelligence is applied in manufacturing.

- A) In writing the payroll of the factory staff
- B) In storing the production records securely
- C) In visual inspection and in predicting equipment failure
- D) In printing the labels for finished goods

Q37. Explain why the success criteria of an Artificial Intelligence project should be agreed before development begins.

- A) They are required by the programming language
- B) They fix the branching factor of the search
- C) They decide what is built and how it will be judged
- D) They determine which sector the system serves

Q38. Interpret the claim that a system belongs to the self-aware class of the functionality classification.

- A) It is the class that every modern system occupies
- B) It is a theoretical class that no system occupies
- C) It means the system keeps a log of its own actions
- D) It means the system can be switched off remotely

Q39. Distinguish an irrecoverable problem from a recoverable one.

- A) A recoverable problem allows no action to be undone
- B) Both of the two allow every action to be undone
- C) Neither of the two allows any action to be undone
- D) An irrecoverable problem allows no action to be undone

Q40. Predict what happens to a search on an infinite state space when no depth limit is imposed.

- A) It may follow one branch forever and never terminate
- B) It reports that no solution exists at all
- C) It is guaranteed to find the shallowest solution
- D) It automatically switches to a heuristic method

----- End of Sample Question Set -----